Citation Integrity Audit

Paper audited: Data Leakage Risks in AI-Assisted Feature Engineering Pipelines
Audit basis: the 20 references listed in Section 9 (Complete References) of the research paper.
Audit date: 28 August 2026

1. Audit Scope

Because the original citation-integrity audit from the earlier assignment was not available, this replacement audit was prepared from the research paper currently stored in the project files. The audit checks whether each cited source can be located in an authoritative scholarly or publisher record and whether the bibliographic information in the paper is consistent with that record. It also flags incomplete or malformed entries rather than silently treating them as fully verified.

2. Verification Method

Sources checked included publisher pages, journal pages, conference proceedings, PubMed, institutional research records, DBLP, and other established scholarly indexes. DOI identifiers were used where available. A 'Verified' result means the source identity and the main bibliographic details were corroborated. 'Verified; citation incomplete' means the source exists but the reference in the paper should be corrected before final submission. 'Indirect relevance' means the source exists but its connection to the specific leakage claim should be described carefully.

3. Reference-by-Reference Audit

#	Reference	Status	Audit finding
1	Feurer & Hutter (2019), Hyperparameter Optimization	Verified	Springer record confirms title, authors, year, pages and DOI 10.1007/978-3-030-05318-5_1.
2	Kaufman et al. (2012), Leakage in data mining	Verified	Publisher/institutional records confirm title, authors, journal, volume/issue, article 15 and DOI 10.1145/2382577.2382579.
3	Sasse et al. (2025), Overview of leakage scenarios	Verified	Springer confirms title, authors, Journal of Big Data 12:135 and DOI 10.1186/s40537-025-01193-8.
4	Kapoor & Narayanan (2023), Leakage and the reproducibility crisis	Verified	ScienceDirect/PubMed confirm title, authors, Patterns 4(9), 100804 and DOI 10.1016/j.patter.2023.100804.
5	Hollmann, Müller & Hutter (2023), CAAFE	Verified with DOI note	NeurIPS confirms the paper and authors. The conference record uses NeurIPS proceedings DOI 10.52202/075280-1938; arXiv DOI 10.48550/arXiv.2305.03403 is also valid for the preprint.
6	Moscovich & Rosset (2022), On the Cross-Validation Bias	Verified	Oxford Academic confirms title, authors, journal, volume 84 issue 4, pages 1474–1502 and DOI 10.1111/rssb.12537.
7	Kanter & Veeramachaneni (2015), Deep Feature Synthesis	Verified	IEEE-indexed records confirm title, authors, DSAA 2015, pages 1–10 and DOI 10.1109/DSAA.2015.7344858.
8	Seyyed-Kalantari et al. (2021), Underdiagnosis bias	Verified; indirect relevance	Nature Medicine bibliographic identity is consistent with the cited DOI. The paper is about hidden/underdiagnosis bias rather than data leakage, so its topical connection should be described narrowly.
9	LLM-Guided Automated Feature Engineering for Time Series Data with Temporal Leakage Control (2026)	Verified; citation incomplete	MDPI confirms title, AI 7(7), article 245 and DOI 10.3390/ai7070245. The manuscript reference omits the author list.
10	Rosenblatt et al. (2024), Data leakage inflates prediction performance	Verified	Nature Communications/PubMed confirm title, authors, volume 15, article 1829 and DOI 10.1038/s41467-024-46150-w.
11	Ichwani et al. (2026), Preventing Data Leakage in Classification	Verified	Jurnal Teknik Informatika confirms title, five authors and DOI 10.52436/1.jutif.2026.7.1.5490.
12	Bouke & Abdullah (2023), Pattern leakage impact	Verified	ScienceDirect/DBLP confirm title, authors, Expert Systems with Applications 230, 120715 and DOI 10.1016/j.eswa.2023.120715.
13	Sasse et al. (2023), On Leakage in Machine Learning Pipelines	Verified	DBLP confirms the arXiv record, title, author list and arXiv identifier 2311.04179.
14	REFORMS (2024), Consensus-based Recommendations	Verified; citation incomplete	Princeton/PubMed confirm the paper. Correct bibliographic details include Science Advances 10(18):eadk3452 and DOI 10.1126/sciadv.adk3452; the manuscript reference omits key author and publication details.
15	Mishra et al. (2026), Taxonomy for temporal data leakage	Verified	PLOS ONE confirms title, four authors, publication date, volume 21(5), article e0340167 and DOI 10.1371/journal.pone.0340167.

#	Reference	Status	Audit finding
16	Hidden Stratification Causes Clinically Meaningful Failures (2020)	Verified; author citation needs correction	The cited arXiv 1909.12475 exists. The manuscript entry is malformed ('Pirsiavash et al. / Oakden-Rayner...'). The paper authors are Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro and Christopher Ré.
17	Jain & Zongker (1997), Feature selection	Verified	Bibliographic records confirm title, authors, IEEE TPAMI 19(2), pages 153–158 and DOI 10.1109/34.574797.
18	Abdi & Williams (2010), Principal component analysis	Verified	Wiley confirms title, authors, WIREs Computational Statistics 2(4), pages 433–459 and DOI 10.1002/wics.101.
19	Khalid, Khalil & Nasreen (2014), Feature selection/extraction survey	Verified	Bibliographic records confirm title, authors, 2014 Science and Information Conference and DOI 10.1109/SAI.2014.6918213.
20	Zheng & Casari (2018), Feature Engineering for Machine Learning	Verified bibliographic identity	The book citation and ISBN are consistent with the published O'Reilly title. This is a book rather than a peer-reviewed journal/conference paper, so it should be labeled as a technical/book source if required by the assignment.

4. Findings Requiring Attention

• **Reference [5]:** The paper identifies CAAFE with an arXiv DOI. The NeurIPS proceedings record also exists and should be preferred when citing the published conference paper.

• **Reference [8]:** The source is a real Nature Medicine paper, but it is primarily about underdiagnosis/hidden stratification rather than data leakage. Any claim should not imply that the paper directly studies leakage.

• **Reference [9]:** The 2026 temporal-leakage paper is real and directly relevant, but the paper's reference entry omits the authors. The entry should be completed.

• **Reference [14]:** REFORMS is a real Science Advances paper. The existing entry is incomplete. It should include the full author list, Science Advances volume/issue, article number eadk3452, and DOI 10.1126/sciadv.adk3452.

• **Reference [16]:** The existing author field is malformed. The cited paper is the 2020 CHIL work 'Hidden Stratification Causes Clinically Meaningful Failures in Machine Learning for Medical Imaging' by Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré.

• **Reference [20]:** This is a book/technical source, not a journal or conference paper. It is acceptable as background if the assignment permits books, but it should not be described as a peer-reviewed research paper.

5. Overall Citation Integrity Assessment

The audit found that the 20 reference entries correspond to identifiable scholarly or technical sources. Most DOI-bearing entries are supported by authoritative records. However, several entries should be corrected for completeness or precision before the repository is presented as a polished research artifact. In particular, references [9], [14], and [16] need bibliographic correction, while [8] and [20] require careful description of their type and relevance.

6. Recommended Corrections

• [9] Add the complete author list for the 2026 AI article.

• [14] Replace the shortened REFORMS entry with the complete Science Advances citation and DOI.

• [16] Replace the malformed author string with the four correct authors of arXiv:1909.12475.

• [8] Keep the source only where its discussion of underdiagnosis/hidden stratification is genuinely relevant; do not present it as a direct leakage study.

• [20] Label it as a book/technical reference rather than a scholarly paper.

• Keep DOI links in the final reference list wherever possible to improve traceability.

7. Audit Conclusion

Citation integrity is **substantially supported but requires minor bibliographic corrections**. The underlying sources are traceable, and the main leakage-related claims are supported by identifiable scholarly literature. The flagged entries should be corrected before the repository is submitted. This audit is intended to document the verification process and does not replace the original academic citation-checking work if the course specifically requires that earlier document.

8. Selected Verification Sources

Springer — Hyperparameter Optimization: https://doi.org/10.1007/978-3-030-05318-5_1

Kaufman et al. — Leakage in data mining: https://doi.org/10.1145/2382577.2382579

Springer — Overview of leakage scenarios: https://doi.org/10.1186/s40537-025-01193-8

Patterns — Leakage and the reproducibility crisis: https://doi.org/10.1016/j.patter.2023.100804

NeurIPS — CAAFE: https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c2df4c35cdbee764ebb9e9d0acd5197-Abstract-Conference.html

Oxford Academic — Cross-validation bias: https://doi.org/10.1111/rssb.12537

IEEE/DSAA — Deep Feature Synthesis: https://doi.org/10.1109/DSAA.2015.7344858

Nature Communications — Data leakage inflates prediction performance: https://doi.org/10.1038/s41467-024-46150-w

PLOS ONE — Temporal data leakage taxonomy: https://doi.org/10.1371/journal.pone.0340167

Princeton REFORMS project: https://reforms.cs.princeton.edu/